Soham Manjrekar

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Education

Georgia Institute of Technology Aug. 2024 – Dec. 2025
Master of Science in Electrical Engineering *Thesis Advisor: Dr. Deepak Divan*

- Chair, IEEE Power & Energy Society Georgia Tech Student Chapter (IEEE Region 3)

University of Illinois Urbana-Champaign Aug. 2020 – May 2024
Bachelor of Science in Electrical Engineering

Research and Work Experience

Power Electronics Engineer Oct. 2025 – Present
GridTran - Endeavour (Founding Engineer) *Atlanta, GA*

- Designing a 98.6% efficient, 5 MW, 34.5 kVac – 800 Vdc Medium-Voltage Solid-State Transformer (MVSST) for a beachhead market of datacenter power delivery
- Co-led Electrical, Mechanical, and Firmware teams through iterative design, validation, and test sprints, rapidly de-risking core technology prototypes in preparation for a New Product Introduction (NPI) line

Graduate Research Assistant Aug. 2024 – Oct. 2025
Georgia Institute of Technology – Center for Distributed Energy (CDE) *Atlanta, GA*

- Research Advisor: Dr. Deepak Divan
- Research focused on designing, modeling and prototyping grid-connected inverters which integrate distributed energy resources to be rapidly adopted and scaled for industrial use

Undergraduate Research Assistant Aug. 2023 – May 2024
University of Illinois Urbana-Champaign – Power and Energy Group *Champaign, IL*

- Advisor: Dr. Arijit Banerjee
- Research to experimentally characterize transformer core losses when excited by a signal of multiple frequencies

Electrical Engineering Intern May 2023 – Aug. 2023
Northrop Grumman - Mission Systems *Baltimore, MD*

- Supported design and analysis of the Power Processing Unit (PPU) for electrical systems on the F-35 Lightning II

Selected Projects

Regenerative Clamp Converter - *Thesis Dissertation*
Center for Distributed Energy (CDE), Georgia Institute of Technology *Atlanta, GA*

- Design and analysis of an active clamping circuit to enable higher power processing in single-stage, inertia-less, isolated converters by capturing and recovering trapped energy in the circuit
- 400 V_{DC} to 600 $V_{AC,rms}$, 3.6 kW hardware prototype, simulated at 80 kW; reduced losses by >90%

Medium Voltage Inertia-Less Isolated Converter
Center for Distributed Energy (CDE), Georgia Institute of Technology *Atlanta, GA*

- Designed, prototyped, and validated a multi-level, single-stage, inertia-less isolated converter
- 800 V_{dc} to 2.4 $kV_{AC,pk}$ per level, rated for 250 kW

Integrated Home Energy Fast Charger
Center for Distributed Energy (CDE), Georgia Institute of Technology *Atlanta, GA*

- Design of a bidirectional, multi-stage, isolated home backup and EV charger with automatic PV/storage integration
- 250–400 V_{DC} to 120 V_{AC} split-phase, rated for 50 kW

Bidirectional, Single-Stage, Inertia-less, Isolated DC/AC Energy Router
Center for Distributed Energy (CDE), Georgia Institute of Technology *Atlanta, GA*

- Designed and prototyped a bidirectional, single-stage, inertia-less, isolated DC/AC energy router with a Σ-Δ pulse-density modulation scheme for islanded, off-grid applications
- 48 V_{DC} to 240 V_{AC}, rated for 1 kW

Publications & Patents

[1] **Soham Manjrekar**, Poornachandra Rao Nallamatti, Joseph Benzaquen, and Deepak Divan. "A Regenerative Clamp Converter for Energy Capture and Recovery in High-Power Single-Stage Isolated Converters". *IEEE Applied Power Electronics Conference & Exposition (APEC)*. Accepted/Selected for Dialogue Preview Session (DPS), March 2027.

[2] **Soham Manjrekar**, Poornachandra Rao Nallamatti, Joseph Benzaquen, and Deepak Divan. "Soft Switching Isolated Inertia-less Modular Multilevel (SSIIMM) Converter". *IEEE Transactions on Power Electronics*. In Review, November, 2026.

[3] **Soham Manjrekar**, Poornachandra Rao Nallamatti, Joseph Benzaquen, and Deepak Divan. "Comparative Analysis of Transformer Saturation in High-Frequency Single-Stage Inertia-less Isolated Converters under PWM and Pulse-Density Modulation". *IEEE Transactions on Power Electronics*. In Review, November, 2026.

[4] Pranjal M. Gajare, Zexian Zeng, Aniruddh Marellapudi, **Soham Manjrekar**, Joseph Sugijanto, Joseph Benzaquen, Maryam Saeedifard, and Deepak M. Divan. "Impact of Switching Nonidealities and Inner Loops on the Dynamics of Grid-Forming Inverters". *IEEE Transactions on Power Electronics*. In Review, November, 2026.

[5] Deepak Divan, Joseph Benzaquen, Poornachandra Rao Nallamatti, and **Soham Manjrekar**. *Soft Switching Isolated Inertia-less Modular Multilevel (SSIIMM) Converter*. Provisional Patent Disclosure, GridTran. March, 2026.

[6] Ruomu Hao, **Soham Manjrekar**, Satish Belkhode, Aniruddh Marellapudi, Joseph Benzaquen, and Deepak Divan. "A Multiport Bidirectional HF-Link Split-phase DC/DC/AC Universal Minimal Converter". *IEEE Energy Conversion Congress & Expo (ECCE)*. October, 2025. DOI: 10.1109/ECCE58356.2025.11259542.

[7] Deepak Divan, Joseph Benzaquen, and **Soham Manjrekar**. *Automatic Integration of PV Solar Energy into Multiport Dual Active Bridge Converter-Based EV Chargers*. Provisional Patent Disclosure, Georgia Institute of Technology. June, 2025.

[8] **Soham Manjrekar**, Aniruddh Marellapudi, Satish Belkhode, Joseph Benzaquen, and Deepak Divan. "AC Cube: A Single-Stage PV/Battery/Grid Energy Router". *2025 IEEE Energy Conversion Congress & Exposition Asia (ECCE-Asia)*. May, 2025, pp. 1–6. DOI: 10.1109/ECCE-Asia63110.2025.11111930.

[9] Navami Prabhu, **Soham Manjrekar**, Ritvik Sriram, Aniruddh Marellapudi, Joseph Benzaquen, and Deepak Divan. "Solar Plug – Universal Off-Grid Microconverter for Low-Cost Tier-1-4 Energy Access". *IEEE International Decentralized Energy Access Solutions (IDEAS) Conference*. January, 2025. DOI: 10.1109/IDEAS62193.2025.11157522.

Awards

Frank C. Mock Scholarship Dec. 2023
Grainger Power Engineering Award May 2024

Skills and Proficiencies

Modeling & Simulation: MATLAB/Simulink, PLECS, PSCAD, LTspice; hardware-in-the-loop (HIL) with ETAS LABCAR and OPAL-RT
PCB & Mechanical CAD: Cadence Allegro PCB Designer and OrCAD schematic capture, Altium 365, KiCad, EAGLE; SolidWorks, 3DEXPERIENCE, Autodesk Fusion 360, OpenSCAD, Blender; rapid additive (FFF, SLA) and subtractive (CNC milling, laser cutting) manufacturing
Power-Stage Testing: Hard- and soft-switching characterization (double-pulse testing); high-frequency transformer and inductor characterization; calorimetric efficiency; thermal runaway testing; back-to-back converter power validation
Insulation Testing: Hi-Pot, basic insulation level (BIL), partial discharge (PD), surge and impulse testing
Embedded Controls: TI C2000 DSP platform (C/C++, Code Composer Studio); SystemVerilog on Intel (Altera) FPGAs (Quartus Prime); Python, Swift
Prototyping & Manufacturing: Soldering and ultrasonic welding; DFM/DFT-driven PCB design and laminate selection (Tg, CTI, thermal and insulation performance); automated manufacturing file creation, BOM management, PCBA RFQs and fabrication EQs